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Primer

Synthetic mechanobiology

John T. Ngo

Cells do not simply endure mechanical forces — they generate, transmit, and interpret them as biological signals. From membrane tension and receptor pulling to substrate stiffness and fluid shear, physical inputs shape migration, differentiation, immune function, and tissue organization. Whereas mechanobiology seeks to understand how living systems read these cues, synthetic mechanobiology treats mechanoregulatory pathways as an input–output relationship that can be deliberately programmed. Integrating tools from synthetic biology with mechanobiological principles, this emerging field reframes force-sensing as a design problem in which mechanical signaling is rewired to achieve useful outcomes — and in which the act of rewiring can itself serve as a powerful investigational strategy. Here, I survey the molecular mechanisms of mechanosensing, describe their use as modular mechanogenetic parts, and discuss applications and open challenges. The long-term vision is to build engineered cells that not only read tissue mechanics but act on them, turning the physical signatures of disease into precise, locally delivered therapeutic programs.

Mechanical inputs, biological outputs

Decades of physiological and cell-biological research have shown that nature's signaling repertoire extends well beyond molecules and photons: mechanical forces — stretch, contraction, bending, shear, and tension — are ubiquitous inputs and outputs in living systems. Mechanobiology has made substantial progress in uncovering how cells sense these forces and translate them into biochemical, cytoskeletal, transcriptional, and mechanical outputs. Synthetic biology, by contrast, treats living systems as engineerable platforms in which defined inputs are transformed into specified outputs through genetic circuits and modular parts. Synthetic mechanobiology sits at the intersection of these fields, not only describing how cells respond to force, but reconstructing and exploiting their mechanosensing and force-generating capabilities so that cellular decisions become both predictable and programmable.

A helpful way to think about this approach is to view cells as biomechanical devices. Like any functional system — whether a smartphone or an engineered tissue — cells integrate three core operations: *sense*, *compute*, and *actuate*. Cells continuously read mechanical cues from their environment, process them through signaling networks, and execute context-dependent outputs. Biomechanical environments, however, present a challenging engineering arena, with forces that are noisy and heterogeneous, and varying in magnitude, direction, loading rate, and geometry. How does a cell distinguish a weak tug from a sustained pull, or a soft matrix from a stiff one, and route each toward an appropriate biological outcome?

From a synthetic perspective, this complexity is both a challenge and an opportunity. Engineered circuits that filter noise, impose thresholds, or sharpen force discrimination can serve not only as tools for controlling cell behavior but also as experimentally tractable models to uncover how natural mechanotransduction encodes and directs the processing of physical information. In the sections that follow, I survey the molecular building blocks of mechanosensing, describe their emerging use as modular mechanogenetic parts, and outline key applications and open challenges.

Sensing: how cells detect force

At the molecular level, mechanosensing arises because forces alter the three-dimensional structures of biomolecules and their assemblies. Applied force can induce conformational changes that modify binding interactions, expose cryptic sites, alter channel properties, and trigger assembly or disassembly. These structural changes activate signaling pathways operating across timescales — from rapid ion fluxes to long-term changes in gene expression and cell behavior.

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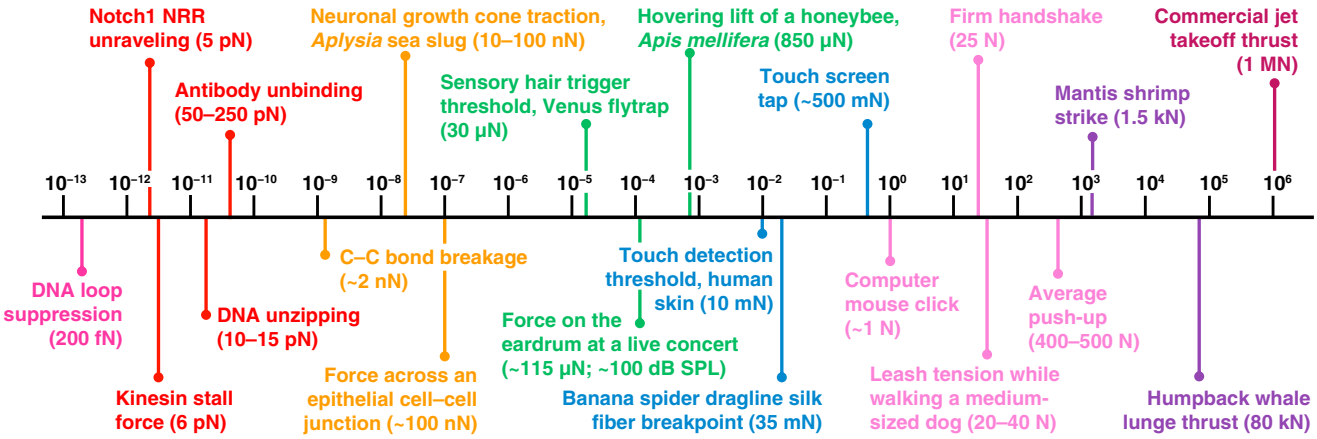


Figure 1. A force ladder from molecules to machines.
Representative forces spanning ~19 orders of magnitude — from single-molecule mechanics (Notch unraveling, molecular motor stalling) through cellular traction, organismal biomechanics (Venus flytrap, spider silk), and human-scale interactions (touchscreen taps, handshake), up to engineered systems (jet thrust). Values are illustrative and context dependent.

Proteins mediating these responses encode mechanical sensitivity in their sequences and structures. Single-molecule force spectroscopy shows that unfolding or rearranging protein domains typically requires forces ranging from a few to several tens of picoNewtons (pN) — force scales matching those generated by molecular motors, cytoskeletal tension, and cell–cell or cell–matrix interactions. As a result, many molecular pathways appear to have incorporated mechanoregulatory processes into their everyday operation: Notch receptors activate in response to endocytic pulling forces, fibronectin unfolds under tension to regulate matrix assembly, and proteasomes act as ATP-fueled motors that mechanically unfold proteins during degradation.

While individual proteins respond to pN-scale forces, cells must interpret mechanical signals spanning much broader ranges (Figure 1) — from molecular-scale protein unfolding to nanoNewton cellular tractions to kiloPascal (N/m^2) tissue stiffnesses. How mechanical inputs are interpreted therefore depends not on force magnitude alone, but on the sensing architecture and how it couples mechanically to its environment. Understanding how distinct sensors operate over specific force regimes is essential for engineering cells that compute with mechanics.

A central challenge is connecting pN-scale conformational transitions to bulk tissue mechanics. Progress will require quantitative approaches that link molecular state, force transmission, and downstream signaling across matched spatial and temporal scales. Synthetic mechanobiology provides a build-to-learn framework for this problem, in which engineered systems with defined force–response properties reveal how molecular-scale mechanical events propagate to cellular outputs and tissue-level properties.

Mechanosensitive ion channels: fast sensors for rapid computation

Mechanosensitive ion channels open or close in response to forces such as stretch, shear, or curvature, thereby gating ion flux across membranes (often including Ca^{2+}, Na^+, K^+, and in some cases Cl^-). Found across all domains of life, mechanosensitive ion channels likely emerged early in evolution. In bacteria, low salt conditions alter membrane tension to impact channels such as MscS and MscL, protecting cells from lysis during hypoosmotic shock. In plants, related channels contribute to touch responses and growth regulation. In animals, mechanosensitive ion channels underlie diverse physiological processes, including touch, hearing, the sense of body position and movement, and even the urge to urinate. Channels formed by Piezo1 and Piezo2 convert mechanical stimuli into cation influxes that initiate downstream signaling (Figure 2A). Loss-of-function mutations in these channels cause a variety of human diseases, underscoring their physiological importance.

Ion channels provide the fastest mechanotransduction timescales — on the order of milliseconds — making them ideal input modules for synthetic circuits requiring rapid mechanical computation. Their sensitivity to membrane-level forces and compatibility with remote actuation modalities position them as potentially versatile components for engineered mechanogenetic systems. For example, ultrasound-mediated activation of Piezo1 has been used to control gene expression in engineered T cells, demonstrating the potential for remote, non-invasive actuation. Ultrasound-targeted, microbubble-mediated mechanoactivation of Ca^{2+} influx activates the phosphatase calcineurin, which in turn then dephosphorylates nuclear factor of activated T cells (NFAT), facilitating its nuclear translocation. This pathway enables ultrasound to act as a programmable input for chimeric antigen receptor (CAR) expression and cytotoxic activity.

TRPV4, a mechanosensitive cation channel naturally responsive to osmotic stress and mechanical strain, illustrates a complementary strategy: in engineered cartilage systems, TRPV4 responds to compression-induced osmotic fluctuations, activating NF-κB signaling and enabling mechanically triggered production of anti-inflammatory proteins, thereby facilitating autonomous,

load-responsive drug delivery. Point mutations in TRPV4 can tune the magnitude and temporal dynamics of circuit output, demonstrating that channel mechanosensitivity can be genetically adjusted to match therapeutic requirements. More broadly, mechanosensitive channels offer potential as sensors to discriminate specific mechanical stimuli (such as fluid shear versus membrane stretch) or their magnitudes, enabling cells to make decisions based on the type and intensity of distinct mechanical signals.

The rapid kinetics of channel gating make ion channels well suited for circuits requiring fast actuation or discrimination between transient noise and sustained mechanical inputs. Because many designs typically funnel through endogenous Ca^{2+} signaling, they can be powerful but are not fully orthogonal and may be prone to crosstalk with native pathways. A useful next step is to design AND-gated systems in which mechanically induced Ca^{2+} fluxes must act together with a second input, thereby improving specificity by restricting actuation to conditions where both signals are present. This form of AND gating is embedded within ultrasound-induced CAR expression, where an initial Ca^{2+} influx is coupled to the antigen specificity of the subsequently expressed receptor (yielding an effective two-input gate). More elaborate AND-gating schemes could further sharpen selectivity, particularly when payload specificity is limited.

Similarly, recent work shows that Piezo1 phosphorylation via protein kinase A and protein kinase C pathways can potentiate mechanosensitivity, with phosphorylated channels displaying slower deactivation kinetics. These findings underscore the complexity of endogenous Piezo regulation and point to potential synthetic applications for mutant or phosphomimetic Piezo variants, for engineering resistance to kinases or for installing 'built-in' potentiation, respectively. In a broader sense, they suggest the possibility of engineering synthetic components whose mechanical responses are modulated through post-translational modifications, thereby adding a regulatory layer atop sequence-encoded mechanosensitivity. In principle, such designs could enable researchers to couple force-sensing behavior to defined signaling states or cellular contexts.

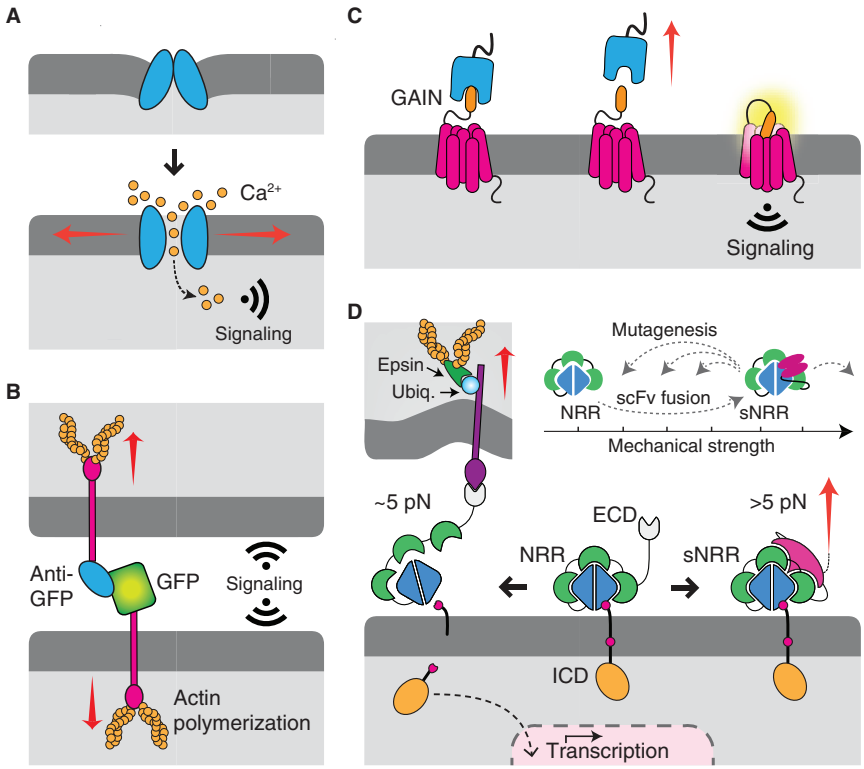


Figure 2. Mechanosensitive receptor architectures.
(A) Piezo channels (blue) convert membrane tension into millisecond-scale calcium (yellow) influx through force-induced pore opening. These influxes then go on to activate downstream signaling by activating Ca^{2+}-dependent effector proteins, such as calmodulin, calcineurin, and calcium/calmodulin-dependent protein kinase II (CaMKII). (B) Adhesion receptors (integrins, cadherins) create bidirectional mechanical linkages; force triggers mechanochemical reinforcement through recruitment of polymerizing actin filaments (yellow), facilitating force-dependent adhesion strengthening. Synthetic cell adhesion molecules ('SynCAMs') enable programmable cell–cell adhesions via orthogonal extracellular binding domains (blue and green). (C) Adhesion GPCRs (aGPCRs) contain GAIN domains that sequester tethered agonists (yellow); force promotes amino-terminal fragment (blue) displacement, exposing the agonist to autoactivate G-protein signaling. (D) Notch operates as a mechanochemical AND gate: activation requires ligand binding plus tensile force to expose proteolytic sites in the negative regulatory region (NRR, blue and green). Tension application is mediated via ligand (purple) binding to the receptor extracellular domain (ECD, gray) and subsequent endocytosis of receptor-bound ligands. Ubiquitination (pale blue) of ligand intracellular tails recruits the endocytic adaptor epsin (green), which mediates coupling to endocytic and force-generating machinery (top left). Fusion of the NRR with a single chain variable fragment (scFv, pink) derived from an NRR-stabilizing antibody results in a 'strengthened NRR' (sNRR) domain, which requires elevated tension for activation (right). Introducing mutations results in sNRR domains with varied mechanical stabilities (~5–50 pN) (top). Activation of Notch, SynNotch, and 'tension-tuned' receptors results in intracellular domain (ICD, yellow) nuclear translocation and downstream target gene transcription.

Adhesion receptors: from natural mechanosensing to synthetic design

Integrins exemplify mechanosensitive adhesion receptors. These heterodimeric transmembrane proteins bind extracellular matrix ligands, such as fibronectin and collagen, while coupling to the actin cytoskeleton through adapter proteins, including talin and vinculin. Force applied across integrin bonds triggers conformational changes that strengthen both ligand binding and cytoskeletal linkages, a phenomenon termed mechanochemical reinforcement. Talin's well described force-dependent regulation provides a prime example of conformational regulation: under tension, its rod domain

unfolds to expose cryptic binding sites that recruit vinculin and reinforce the integrin–cytoskeleton connection. This force-dependent unfolding, demonstrated by single-molecule force spectroscopy and validated in living cells, represents a generalizable mechanism whereby mechanical force gates biochemical signaling.

While integrins mediate cell–matrix adhesion, cadherins establish mechanical linkages between cells. Like integrins, cadherins couple to the actin cytoskeleton and exhibit force-dependent strengthening. Adhesion receptors thus function as bidirectional mechanical links capable of both generating and responding to forces across cellular boundaries. Their modular architecture — extracellular binding domains, transmembrane segments, and cytoplasmic scaffolds — creates force-bearing pathways that can be engineered to transmit, amplify, or gate mechanical signals. Recent synthetic cadherin variants enable programmable cell–cell recognition, offering tools to engineer cell adhesion and tissue organization based on mechanical as well as orthogonal biochemical cues (Figure 2B). Experiments with these synthetic cell adhesion molecules show that cytoskeletal coupling, rather than receptor–ligand binding interactions, governs adhesive outcomes.

Beyond integrins and cadherins, adhesion G-protein-coupled receptors (aGPCRs) couple adhesive extracellular domains to seven-transmembrane signaling cores. These receptors contain a GAIN domain that undergoes autoproteolysis to generate a non-covalent complex in which a tethered agonist peptide is sequestered within the domain's three-dimensional fold. Mechanical force can promote displacement of the amino-terminal GAIN fragment, exposing the cryptic agonist, which then autoactivates the aGPCR by binding its ligand-binding site, thereby triggering downstream G-protein signaling (Figure 2C). This architecture makes aGPCRs attractive scaffolds for synthetic mechanobiology: extracellular domains define where forces are sensed, GAIN domain displacement converts force to agonist exposure, and intracellular coupling can be rewired to orthogonal outputs. Such designs complement recently developed synthetic GPCR strategies that decouple activation from downstream signaling responses.

Force-gated receptors: Notch and SynNotch systems

Force-gated receptors provide a powerful strategy for mechanotransduction. In canonical Notch signaling, ligand binding alone is insufficient for activation, with productive signaling requiring tensile force across the receptor–ligand complex — a mechanism likely established to prevent leaky signaling. This force exposes cryptic proteolytic sites within the receptor's negative regulatory region (NRR), enabling cleavage and release of the Notch intracellular domain, which then translocates to the nucleus to regulate downstream gene expression (Figure 2D). In the device framework, Notch functions as an irreversible, mechanochemical 'AND' gate, integrating ligand engagement with force transmission during cell–cell contact. Even though synthetic Notch (SynNotch) receptors leverage endogenous proteolysis, their ligand-binding module and released transcription factor are user-definable, thus enabling end-to-end orthogonality from ligand to output, providing an advantage compared to channel-based systems that route signals through shared second messengers.

Notch's activation threshold is encoded structurally. The NRR acts as a mechanical safety catch, preventing cleavage until force exceeds a defined threshold and thereby suppressing activation by weak or transient interactions. Importantly, this force requirement varies across species, with comparative studies revealing that Notch receptors have been naturally tuned through evolution to operate over distinct mechanical regimes. For example, whereas *Drosophila* and vertebrate Notch require force generated by ligand endocytosis, *Caenorhabditis elegans* Notch can be activated with lower force due to subtle structural differences in the NRR that reduce mechanical shielding of the so-called 'S2' cleavage site. These observations demonstrate that evolution modulates Notch force sensitivity by adjusting protein architecture.

SynNotch receptors exploit this force-gated architecture by decoupling ligand recognition from downstream transcriptional outputs. By preserving force-dependent proteolysis while modularizing inputs and outputs, SynNotch enables engineered cells to activate user-defined gene programs in response to mechanically productive interactions.

Recent synthetic implementations extend this principle by explicitly tuning mechanical sensitivity. By engineering structural elements within force-bearing regions, synthetic receptors can be programmed to activate over defined force ranges, transforming mechanical force from a binary trigger into a selective or graded input. Natural Notch receptors and tension-tuned SynNotch systems together illustrate a central principle of synthetic mechanobiology: mechanosensitivity is an evolvable and engineerable property encoded in protein sequence and structure. Consistent with this view, experiments have shown that even single amino acid changes can markedly alter how proteins sense and transmit mechanical signals, raising the biological question of why evolution has tuned proteins to particular force thresholds. Simultaneously, these findings show that force thresholds can be redesigned and exploited as tunable circuit parameters.

Force threshold tuning may be only part of the circuit. In Notch signaling, catch bonding between Notch1 and its ligands may help commit trans-cellular complexes to productive signal generation, preventing slippage as intercellular forces increase toward an optimum. By analogy, this resembles a handshake where the grip tightens as one party pulls inward, effectively enforcing signal transmission to receiver cells rather than inviting its reception. Beyond its optimum, Notch–ligand bond lifetime decreases (catch–slip behavior), perhaps creating a natural mechanical 'filter' that ensures signaling specificity in response to endocytic forces (rather than other forces that might exceed endosomal magnitudes).

To date, studies of Notch catch bonding have largely focused on human Notch1–ligand interactions, and it remains to be seen whether catch bonding is a Notch1-specific

feature or a more general property that extends to other Notch paralogs (Notch2–4). If catch-bond behavior differs across receptor–ligand pairs, or is absent for some paralogs, these mechanochemical differences could help explain why particular ligands preferentially activate particular Notch receptors. Conversely, if Notch2–4 receptors lack catch-bond behavior, this would suggest the equally interesting possibility that force-stabilized signaling is a Notch1-specific adaptation rather than a conserved feature of the pathway. Notch is expressed broadly across tissue types, and catch bonding may help maintain signaling specificity to ligand endocytosis across heterogeneous mechanical landscapes. Comparative and evolutionary analyses of ligand and receptor orthologs may provide complementary insights, revealing not only when force-stabilized Notch–ligand binding emerged, but also hinting at the selective pressures that may have shaped such mechanistic tuning. Together, these questions underscore how much remains to be learned about natural force-gated systems, while also highlighting how much opportunity exists to engineer new ones.

Force-gated cryptic sites: a generalizable strategy

The principles underlying Notch mechanosensing extend beyond receptor engineering. Force-gated cryptic sites (domains that unfold or expose regulatory sequences only under tension) represent a generalizable strategy for building synthetic mechanosensors. Just as Notch requires force to expose cryptic cleavage sites, synthetic proteins can be designed with mechanically protected domains tailored to specific force ranges, loading rates, or pulling contexts.

An important practical constraint is that force can only be transmitted to these cryptic domains if the upstream ligand–receptor bond is mechanically robust, since, as in any chain, the weakest link determines where rupture occurs. The Bell–Evans model provides a framework for anticipating which bonds will fail first, answering a simple but fundamental question: if a slip bond is held at force F, how quickly does it break? Here, the force-dependent off-rate is given by:

$$k_{off}(F) = k^0_{off} e^{F x_\beta / k_B T}$$

where k^0_{off} is the intrinsic off-rate, x_β is the distance to the transition state, and $k_B T$ is thermal energy. Because $k_{off}(F)$ increases exponentially with force, bonds with lower intrinsic off-rates survive longer under load and can thus transmit tension to downstream elements before rupturing. This principle was illustrated in the development of tension-tuned receptors: strengthened receptors fused with mechanically weak binding domains failed to activate in response to immobilized or cell-presented ligands, whereas activation was restored upon substitution with a mechanically stable (slow intrinsic k_{off}) binding domain. Models like Bell–Evans should prove broadly useful for synthetic mechanobiology. Rupture force measurements remain uncommon, but k_{off} values are widely available; these models can therefore approximate relative mechanostabilities and help guide interaction domain selection even without direct force spectroscopy data.

Computation: extracting meaning from mechanics

Cells and devices must do more than detect signals — they must interpret them. Just as a smartphone distinguishes taps from swipes, or single versus sequential button hits, cells must determine not only whether force is present but also its magnitude, duration, direction, and context. In biological systems, mechanical cues are inherently noisy: forces fluctuate over time, vary in geometry, and differ across microenvironments. Meaningful responses therefore require mechanisms that filter transient fluctuations, impose thresholds, and integrate signals over appropriate timescales.

Natural systems implement mechanical computation through multiple strategies operating across timescales. Rapid Ca^{2+} influx through mechanosensitive ion channels activates signaling pathways that convert transient mechanical inputs into context-specific transcriptional responses. Transcriptional regulators such as YAP/TAZ integrate longer-lived mechanical stimuli, such as substrate stiffness, cytoskeletal tension, and cell shape, to produce graded outputs, while mechanosensitive genomic enhancers (‘mechanoenhancers’) add an additional layer of computation by modulating chromatin accessibility to amplify or stabilize mechanically driven gene expression.

Synthetic mechanobiology builds on these natural principles by making mechanical computation explicit and programmable. Engineered systems can incorporate logic gates, temporal filters, or externally controlled switches to define how mechanical signals are delivered, interpreted, and integrated with biochemical cues. Recent work has demonstrated that cells can be engineered with incoherent feedforward loops (IFFLs, circuit motifs in which an input simultaneously activates a target and a repressor of that target) to express genes selectively at intermediate force levels. In this design, one receptor activates a target gene at low force, while a second, higher-threshold receptor activates a repressor that silences the target at high force, in effect creating a ‘band-pass’ response (Figure 3). Such designs enable cells to discriminate not just whether tension is present, but whether it falls within a specific mechanical range. This synthetic discrimination mirrors natural stem cell mechanosensing, where mesenchymal stem cells differentiate into muscle on intermediate-stiffness substrates while pursuing other fates on softer or stiffer materials. By tuning thresholds and integration rules, synthetic circuits aim to make mechanical decision-making more predictable and portable across diverse cellular contexts.

Actuation: understanding and engineering how mechanical signals drive change

Cells and smartphones must ultimately act on the mechanical information they sense and compute. In a device, the push of a button may silence an alarm, launch an application, or reroute a call. Similarly, mechanical inputs in biological systems drive outputs that reshape cellular behavior, tissue organization, or the surrounding microenvironment.

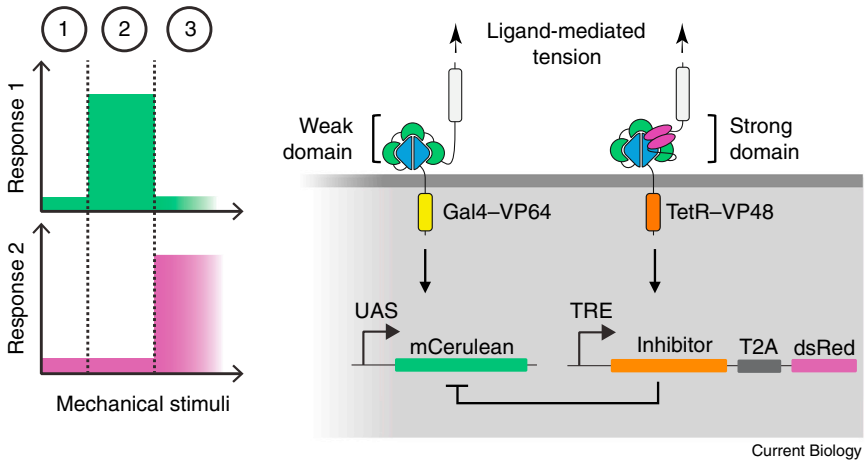


Figure 3. Synthetically designed band-pass filters.
Cells can be engineered to respond selectively to intermediate mechanical forces using an incoherent feedforward loop (IFFL) built from tension-tuned SynNotch receptors. In this design, two SynNotch receptors recognize the same ligand but activate at different force thresholds: a low-threshold receptor senses modest tension and activates a target gene, while a higher-threshold receptor senses stronger tension and induces a transcriptional repressor that suppresses the same gene. By integrating these opposing inputs, the circuit computes whether applied tension falls within an intermediate range. Engineered cells then actuate gene transcription only within this defined mechanical window, remaining quiescent at lower or higher forces. Adapted from Sloas *et al.* (2023), reproduced with permission from SNCSC.

Mechanical signals can elicit genetic actuation, such as the expression of cytoskeletal components, secretion of cytokines, activation of differentiation programs, or, in synthetic contexts, the production of engineered receptors, signaling proteins, or enzymes. These outputs often operate on longer timescales, enabling long-term adaptation to sustained mechanical conditions. Mechanical inputs can also trigger physical actuation, including actomyosin recruitment, localized actin polymerization, focal adhesion growth, membrane remodeling during endocytosis, and changes in cell shape and polarity — responses that can act on faster timescales, alter cell and tissue organization, and directly reshape force generation and transmission. In many settings, genetic and physical actuation are coupled: cytoskeletal remodeling changes tension and adhesion, which in turn changes mechanosensor inputs and transcriptional states.

In synthetic mechanobiology, actuators can be chosen to amplify, buffer, or counteract an initiating mechanical signal. Force-induced expression of cytoskeletal regulators may reinforce adhesion and traction, whereas mechanically triggered secretion of matrix-modifying enzymes could soften or remodel the local environment. Actuation can extend beyond individual cells: engineered cells can export mechanical decisions outward via paracrine cues, matrix remodeling, or contact-mediated interactions, thereby influencing collective behavior in tissues and tissue-like systems.

Many natural designs implement feedback loops, creating cycles of sense → compute → actuate → sense again. Feedback stabilizes behavior, prevents runaway reinforcement, and enables homeostatic control. Negative feedback can dampen excessive force generation, while positive feedback can lock in mechanically specified states such as polarity, lineage commitment, or collective organization. As in engineered devices, robust mechanobiological systems depend on careful integration of sensing, computation, and actuation, with explicit attention to timescales, gain, and coupling between mechanical, electrical, and biochemical processes.

Designing with forces: key principles

Mechanical inputs differ from chemical or optical ones in a crucial way: they are inherently contextual. Force is a vector quantity characterized by both magnitude and direction. The same nominal force can produce distinct outcomes depending on geometry, loading rate, direction, and whether surrounding structures deform or relax. Single-molecule measurements illustrate this sensitivity. Atomic force microscopy studies show that the unfolding force of a 76 amino acid ubiquitin domain depends strongly on pulling direction, while experiments with dihydrofolate reductase demonstrate that identical pulling orientations can result in distinct unfolding forces depending on small-molecule binding. Applied force can also gate substrate recognition, with ~10 pN applied tensions increasing metalloprotease-mediated collagen cleavage by roughly 100-fold. Catch bonds add further complexity: in certain receptor–ligand pairs, such as P-selectin and P-selectin glycoprotein ligand-1, applied force stabilizes binding rather than promoting dissociation. Designing force-responsive systems therefore requires specifying not only the sensor and downstream circuit, but also the mechanical environment and the force–response behaviors of individual components.

Several principles guide this design process. First, mechanosensitivity is genetically tunable. Mechanical responsiveness is encoded in protein structure, so changes such as mutations, domain swaps, circular permutations, or altered stabilizing interactions can shift when and how a protein responds to force. Common architectural features, such as autoinhibitory domains or helical bundles that open under tension, provide concrete ways to tune sensitivity. Antibody–antigen interactions may represent another class of force-sensitive encodable entities, offering an array of building block possibilities for constructing synthetic mechanosensing circuitries. At the single-molecule level, antigen–antibody interactions often fail under forces in the tens-of-picoNewtons regime, and their effective mechanical stability can be tuned further through geometry and affinity (specifically, k_{off}), as well as by monomerization or multivalent presentation.

Second, timescales are critical when designing or interpreting mechanosensitive circuits. Mechanical events span a wide temporal range: ion channels can respond in milliseconds, transcriptional programs may take hours, and tissue-level mechanical properties can change over years. For example, slow extracellular matrix remodeling and collagen crosslinking gradually stiffen tissues such as skin, blood vessels, lungs, and load-bearing joints, altering the baseline mechanical environment that cells experience. Circuits that connect fast mechanical inputs to slower downstream responses must therefore determine whether to react to brief, transient physical stimuli or only to those that persist over time.

Achieving this temporal discrimination poses both an engineering challenge and a core mechanobiology question: do cells already possess mechanisms that allow them to distinguish short-term mechanical signals from long-term changes in tissue stiffness, and if so, how do these mechanisms adapt as tissues age or remodel? Understanding how natural systems integrate mechanical signals across timescales may inform the design of synthetic circuits that remain stable and functional despite slow mechanical drift. Systems aimed at maintaining mechanical homeostasis may also require feedback to avoid overcorrection or oscillatory behavior.

Third, materials are part of the circuit. Viscoelastic substrates may appear soft during rapid loading but stiff under slow deformation, while ligand density and matrix architecture can influence adhesion independently of bulk stiffness. These physical properties act as functional components of the circuit and must be designed alongside signaling elements — in device terms, hardware shapes signals as much as software.

Applications: emerging design patterns

Synthetic mechanobiology intersects multiple areas of bioengineering, offering ways to use mechanical signals as controllable inputs and as informative cues about cellular or tissue states. For example, mechanical gating can sharpen immune recognition by requiring both receptor–ligand binding and productive force transmission, helping discriminate strong interactions from weak or transient ones. Such designs might help engineered immune cells recognize their targets with increased specificity. Synthetic systems could serve as testbeds for exploring these ideas, providing new ways to probe immune mechanotransduction and to design immune responses that depend on mechanical context as well as molecular identity.

Mechanics is not an add-on feature in immunity: it is part of the recognition algorithm. T-cell receptors engage peptides bound to major histocompatibility complex molecules in the context of actin-driven forces at the immunological synapse, and force-dependent bond lifetimes can contribute to kinetic proofreading and antigen discrimination. CARs, built from antibody-derived binding domains, may sample different force–lifetime landscapes and coupling geometries; spacer length, epitope position, and cytoskeletal linkage can therefore shape activation in ways that are not captured by affinity alone.

Recent work has shown that catch bonds can be ‘fished’ from libraries of mutant T-cell receptors, and broadly extending this capability to other binding pair systems remains a key challenge. Newly developed ‘histidine-scanning’ approaches offer a generalizable workflow: scanning for histidine residues identifies interfacial hotspots, which are then randomized into polar/charged residue libraries for functional selection. These findings support a model in which interfacial shearing under load promotes new hydrogen bonds and salt bridges that stabilize force-dependent binding.

Looking ahead, an important goal is to develop sequence–structure rules that define the force windows over which catch bonds engage and the thresholds beyond which they rupture. From a biological standpoint, the forces at which natural catch bonds peak may not be arbitrary; instead, they may reflect the mechanical regimes where their function is most critical — where antigen discrimination, adhesion, or signal activation is most consequential. From a synthetic standpoint, engineered catch bonds will need to operate within similar constraints. Ultimately, their utility will depend on force-lifetime properties that match (or at least meet) the loading rates, force optima, and mechanical contexts of their intended cellular applications.

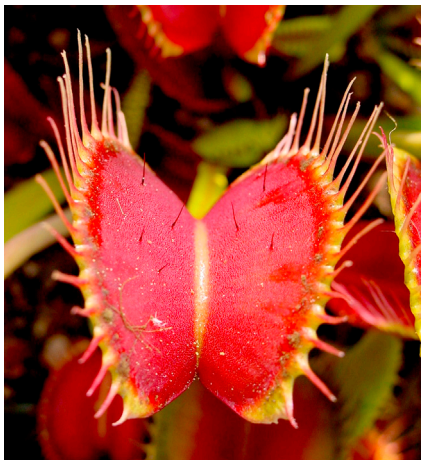


Figure 4. The Venus flytrap as a biomechanical logic gate.
The Venus flytrap (*Dionaea muscipula*) implements a complete sense → compute → actuate cycle. Trigger hairs sense mechanical deflection via mechanosensitive ion channels, generating action potentials. A molecular counting mechanism computes whether two touches occurred within a specific time frame, filtering single-touch noise. Upon threshold crossing, the plant actuates rapid trap closure by releasing stored elastic energy, causing the bilobed leaf to rapidly snap from a convex to a concave shape. This temporal integration allows the flytrap to distinguish real prey from environmental noise such as wind or rain, ensuring that energetically costly closure occurs only when capture is likely. Image by Noah Elhardt/Wikimedia Commons (CC BY-SA 2.5).

Remote actuation adds another dimension: ultrasound can deliver focused mechanical energy deep into tissues. Genetically encoded gas vesicles (air-filled protein nanostructures) can serve as acoustic actuators, converting ultrasound pulses into potent local mechanical effects through inertial cavitation. This enables engineered cells to lyse on command and release therapeutic payloads at targeted sites. Expanding the library of remotely controllable mechanoactuators — through design and discovery — will further broaden the synthetic mechanobiology toolkit. As these approaches mature, an important goal will be to calibrate dose-to-force relationships *in situ*, enabling

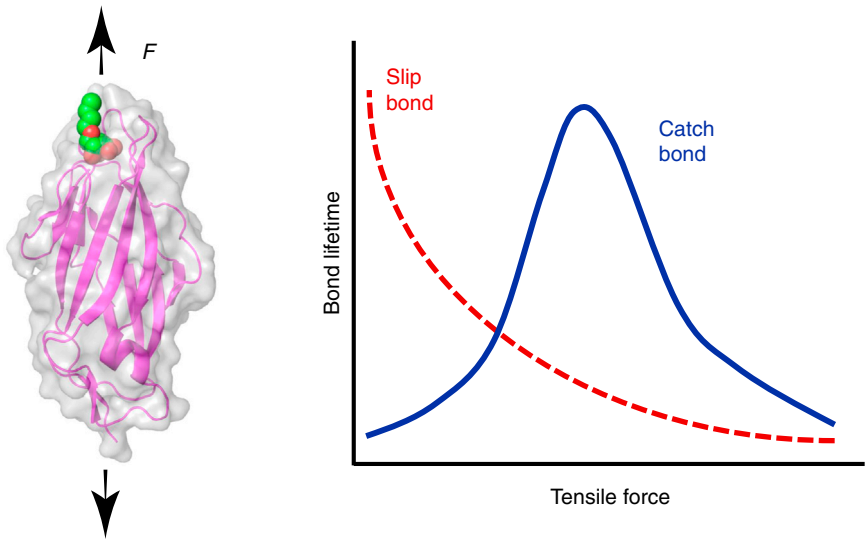


Figure 5. Catch bonds as force-activated molecular switches.
The *E. coli* pilus adhesin FimH (left, PDB: 1UWF) employs catch-bond mechanics to grip host surfaces under shear flow. FimH (magenta) binds mannose groups (green) on receptors expressed on bladder epithelial cell surfaces; at low forces, these bonds dissociate rapidly. Under applied tension, FimH undergoes allosteric rearrangements that shift it from a low-affinity to a high-affinity conformation, extending bond lifetime and reinforcing its adhesion to host cells. This force-induced strengthening allows bacteria to adhere specifically in high-shear environments like the urinary tract. From the biomechanical device perspective, FimH senses shear force and computes whether flow exceeds an adhesion threshold, actuating selective attachment by stabilizing high-affinity bonds. A graph (right) depicts the bond lifetimes of 'slip bonds' (red) versus 'catch bonds' (or 'catch-slip bonds', blue) as a function of tensile force.

ultrasound (or other modalities) to function as a quantitative input rather than a binary trigger.

Disease-associated mechanical changes offer additional targeting opportunities. Patients often first notice something is wrong when they feel a lump, highlighting both the clinical relevance of altered stiffness and our intuitive capacity to sense it. For example, solid tumors can exhibit stiffness values of 10–100 kPa compared to the 0.5–5 kPa values of normal tissues, and fibrotic liver exceeds 20 kPa. Synthetic approaches to link stiffness signatures to therapeutic outputs could enable engineered cells to activate preferentially in these pathological contexts, complementing biochemical targeting when molecular markers are shared but mechanical differences are pronounced.

Synthetic morphogenesis represents an additional study-and-design possibility. Tissues are mechanical as well as genetic systems: as they grow, they fold, bend, and generate forces that help determine form. A synthetic mechanobiology perspective is to treat these forces — and the cellular programs that generate and respond to them — as controllable design variables. By giving cells tunable ways to apply forces or change adhesion, and coupling those actuators to mechanoresponsive sensors, one can build feedback in which cells detect deviations in stress or shape and respond in ways that steer collective organization. In principle, such feedback control could guide robust self-organization, repair, or homeostasis in organoids and engineered tissues, while offering a tractable way to understand how mechanical feedback helps generate and stabilize tissue-scale form.

Seeding precision: delivering defined mechanical stimuli

Engineering mechanosensitive systems requires not only tools for measuring responses but also quantitative methods for delivering defined stimuli reproducibly. Without calibrated inputs, distinguishing engineered phenotypes from noise becomes difficult, and comparing results across laboratories becomes challenging. While this primer has focused mostly on mammalian systems, recent work in plant mechanobiology illustrates both the challenge of controlling stimulus delivery and a solution. Touch, wind, and vibration shape stem thickness, root architecture, and growth across plant species, and plant enthusiasts will know that many carnivorous plants mechanically sense their prey during capture (Figure 4). At the same time, plant synthetic biology aims to advance sustainable agriculture, and, in principle, engineering plant mechanosensing could yield resilient crops that sense and adjust to changing environmental conditions. But in the wild, mechanical signals arrive in heterogeneous and unpredictable forms, whereas in the laboratory, meaningful results demand precision.

In recent work, researchers substituted prey-delivered stimuli with pulsed ultrasound, enabling calibrated force delivery to carnivorous plant mechanosensors and avoiding the variability of live prey. By delivering quantified forces in a controlled manner, they discovered that mutants lacking specific mechanosensitive channels require stronger stimulation to trigger hunting responses. This example, like the use of ultrasound to activate CAR expression described earlier, points to a broader need: as synthetic mechanobiology matures, new tools for reproducibly delivering mechanical stimuli will become increasingly important. These tools will need to span diverse force regimes, timescales, and penetration depths, and accessible methods that can be readily adopted across laboratories will be key to the growth of mechanobiology in both its synthetic and natural forms.

Perspectives

Synthetic mechanobiology reframes mechanical forces as programmable inputs and outputs. By designing systems that sense, integrate, and respond to force in simplified ways, researchers can test mechanistic hypotheses difficult to access through perturbation alone, define minimal components, and characterize critical thresholds

within cellular contexts. While we are far from fully recapitulating natural mechanobiological processes, viewing cells as biomechanical devices offers a productive abstraction — it clarifies how mechanical information flows by separating sensation, computation, and actuation into modular operations. A motivating application is 'mechanical precision' therapy, in which engineered cells detect pathological mechanics and respond locally, only when the physical context is appropriate. In addition to its practical utilities, this framing may sharpen efforts to understand force-responsive systems while also simplifying concepts for classroom teaching.

Yet mechanobiology is highly complex, and our ability to probe and engineer it is in its earliest phase. Current challenges include the limited dynamic range of many mechanosensors, crosstalk between mechanical and biochemical pathways, and difficulties multiplexing orthogonal mechanical inputs within single cells. Progress will require both an improved library of characterized parts (mechanosensors, force-gated receptors, tunable linkers, and catch-bonding pairs) and better frameworks for predicting how these parts behave in deformable, actively remodeling environments. Critically, the development of deployable methods and techniques that can be readily implemented by new users will be key to expanding what we can learn, or what we can design, via synthetic mechanobiological approaches.

Finally, this abstraction is not merely conceptual; biology demonstrates mechanically driven information processing across scales and kingdoms. The Venus flytrap (*Dionaea muscipula*) operates at the organ scale — trigger hairs sense deflection, a counting mechanism integrates inputs over time to filter noise, and stored elastic energy actuates rapid closure (Figure 4). At the molecular scale, bacterial adhesins like *Escherichia coli* FimH exploit catch-bond mechanics, counterintuitively strengthening under tension to maintain attachment in high-shear environments such as urinary tract flow (Figure 5). Where the flytrap uses mechanical energy storage for predation, bacteria employ force-activated adhesion for colonization.

These examples illustrate that evolution tunes biological systems to operate within their mechanical environments, providing synthetic biology with design principles shaped by physical constraints.

DECLARATION OF INTERESTS

The author is a co-founder and stockholder of Motive Therapeutics and an inventor on patents held by the Trustees of Boston University, including one on tension-tuned receptors (U.S. Patent 10,858,443).

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